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Comptia N+ Assignment

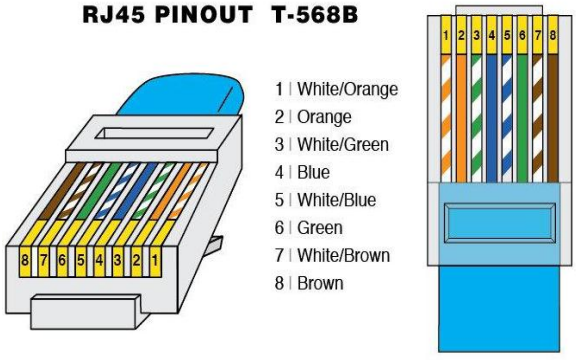
[N+ Network Fundamental and building networks]

Ethernet standard and specification

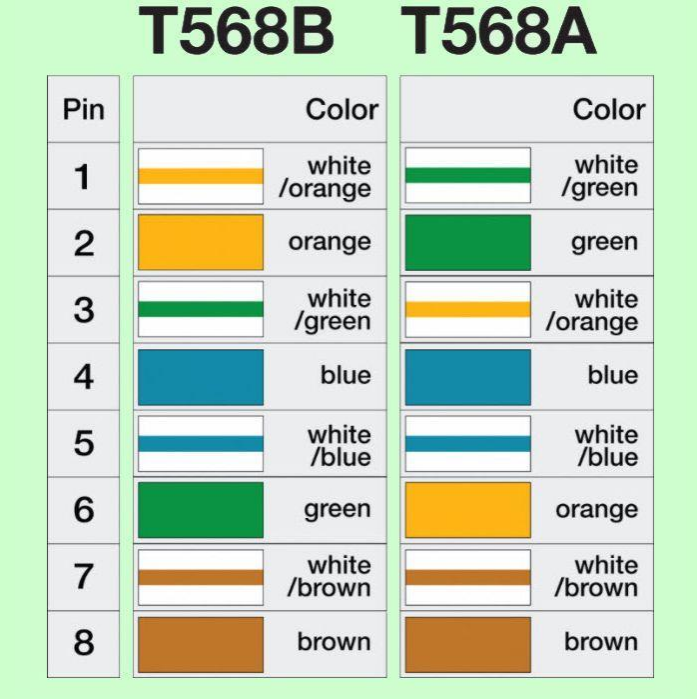
Task:

- **1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.**

  - **Introduction**
    - An RJ-45 connector is commonly used to connect Ethernet cables to networking devices such as computers, routers, and switches. It has eight pins used for transmitting and receiving data. T568A and T568B are two standard wiring schemes that specify the arrangement of wire colours inside the connector.
  - **Diagram**



➢ T568A Wiring Standard and T568B Wiring Standard



➢ Difference Between T568A and T568B

- The main difference between T568A and T568B is the position of the green and orange wire pairs. Both standards are used for Ethernet cable termination, but the same standard should generally be followed at both ends of a straight-through cable.

❖ 2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

➢ Introduction

- Ethernet is a widely used networking technology for connecting computers, routers, switches, and other devices in a Local Area Network (LAN). Different Ethernet standards provide different data transmission speeds and cable specifications.

➢ Comparison Table of Ethernet Standards

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable Type	Cat3 or higher twisted-pair cable	Cat5 or higher twisted-pair cable	Cat5e or higher twisted-pair cable
Maximum Cable Length	100 metres	100 metres	100 metres
Connector Type	RJ-45 (8P8C)	RJ-45 (8P8C)	RJ-45 (8P8C)
Number of Wire Pairs Used	2 pairs	2 pairs	4 pairs
Common Application	Older LANs and basic networking	Office networks and older LANs	Modern LANs, computers, and high-speed networks

- **Explanation**

1. **10BASE-T:** This Ethernet standard supports a speed of 10 Mbps. It commonly uses twisted-pair copper cable and is mainly found in older networking systems.

2. **100BASE-TX:** Also known as Fast Ethernet, it supports a speed of 100 Mbps. It commonly uses Cat5 twisted-pair cable and provides faster data transmission than 10BASE-T.

3. 1000BASE-T: Also known as Gigabit Ethernet, it supports a speed of 1 Gbps. It uses all four twisted pairs of a Cat5e or higher cable and is widely used in modern computer networks.

❖ **3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why. Think about how transmit and receive pairs are wired in each cable type.**

➢ **Introduction**

- Ethernet cables are used to connect computers and other networking devices for data communication. The two common types of Ethernet cables are straight-through cables and crossover cables. They differ in the arrangement of their transmit and receive wire pairs.

➢ **Suitable Cable: Crossover Cable**

- If two laptops need to be connected directly using Ethernet, a **crossover cable** is traditionally used when the network interfaces do not support Auto-MDI/MDI-X.

➢ **Reason for Using a Crossover Cable**

A crossover cable connects the transmitting pins of one laptop to the receiving pins of the other laptop. This allows both laptops to send and receive data correctly.

In a traditional 10/100 Mbps Ethernet connection, the transmit and receive pairs are crossed as follows:

- Pin 1 of one end connects to Pin 3 of the other end.
- Pin 2 of one end connects to Pin 6 of the other end.
- Pin 3 connects to Pin 1.

- Pin 6 connects to Pin 2.

➢ This arrangement allows the transmit pair of one laptop to connect to the receive pair of the other laptop.

➢ **Difference Between Straight-Through and Crossover Cables**

Feature	**Straight-Through Cable**	**Crossover Cable**
Wiring arrangement	Same standard at both ends, such as T568B–T568B	Different standards at the ends, such as T568A–T568B
Transmit and receive pairs	Pins remain in the same positions	Transmit and receive pairs are crossed
Traditional use	Connecting a laptop to a switch or router	Directly connecting two similar devices
Direct laptop-to-laptop connection	May work with Auto-MDI/MDI-X	Traditionally used when Auto-MDI/MDI-X is unavailable

❖ **4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.**

➢ **Introduction**

- Ethernet is a wired networking technology used to connect devices for reliable and high-speed data communication. Different devices support different Ethernet standards depending on their hardware specifications and network requirements.

➢ **Examples of Devices and Their Ethernet Standards**

Device	Likely Ethernet Standard	Use
Wi-Fi Router	1000BASE-T (Gigabit Ethernet)	Connects to computers, switches, and other wired network devices for high-speed data transmission.
Desktop Computer in a College Computer Lab	1000BASE-T (Gigabit Ethernet)	Connects to the college network for internet access, file sharing, and accessing online resources.
Network Printer in a College Office	100BASE-TX (Fast Ethernet) or 1000BASE-T (Gigabit Ethernet)	Connects to the local network so multiple users can print documents.

- **Explanation**

1. **Wi-Fi Router:** Many modern routers have Gigabit Ethernet ports that support 1000BASE-T. These ports provide high-speed wired connections to computers and other networking devices.

2. **Desktop Computer:** Modern desktop computers commonly support 1000BASE-T through their Ethernet ports. This standard allows fast and reliable communication within a college network.

3. **Network Printer:** A network printer may support 100BASE-TX or 1000BASE-T, depending on its model. It uses an Ethernet connection to receive print jobs from computers connected to the same network.